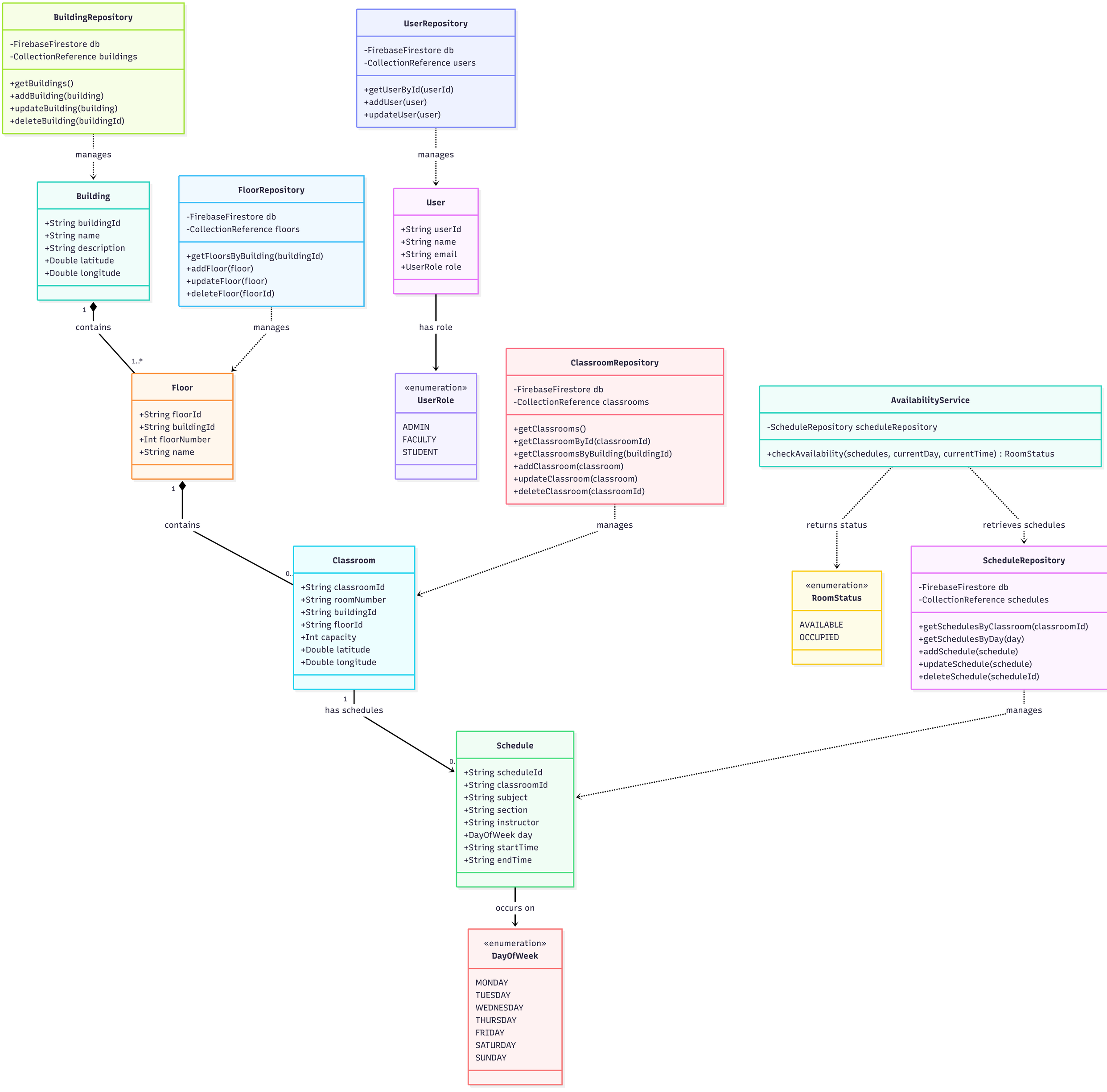

BuildingRepository
-FirebaseFirestore db
-CollectionReference buildings
+getBuildings()
+addBuilding(building)
+updateBuilding(building)
+deleteBuilding(buildingId)
manages
Building
+String buildingId
+String name
+String description
+Double latitude
+Double longitude
1
contains
1..*
FloorRepository
-FirebaseFirestore db
-CollectionReference floors
+getFloorsByBuilding(buildingId)
+addFloor(floor)
+updateFloor(floor)
+deleteFloor(floorId)
manages
Floor
+String floorId
+String buildingId
+Int floorNumber
+String name
1
contains
0..
Classroom
+String classroomId
+String roomNumber
+String buildingId
+String floorId
+Int capacity
+Double latitude
+Double longitude
1
has schedules
0..
Schedule
+String scheduleId
+String classroomId
+String subject
+String section
+String instructor
+DayOfWeek day
+String startTime
+String endTime
occurs on
«enumeration»
DayOfWeek
MONDAY
TUESDAY
WEDNESDAY
THURSDAY
FRIDAY
SATURDAY
SUNDAY
UserRepository
-FirebaseFirestore db
-CollectionReference users
+getUserById(userId)
+addUser(user)
+updateUser(user)
manages
User
+String userId
+String name
+String email
+UserRole role
has role
«enumeration»
UserRole
ADMIN
FACULTY
STUDENT
ClassroomRepository
-FirebaseFirestore db
-CollectionReference classrooms
+getClassrooms()
+getClassroomById(classroomId)
+getClassroomsByBuilding(buildingId)
+addClassroom(classroom)
+updateClassroom(classroom)
+deleteClassroom(classroomId)
manages
«enumeration»
RoomStatus
AVAILABLE
OCCUPIED
AvailabilityService
-ScheduleRepository scheduleRepository
+checkAvailability(schedules, currentDay, currentTime) : RoomStatus
returns status
retrieves schedules
ScheduleRepository
-FirebaseFirestore db
-CollectionReference schedules
+getSchedulesByClassroom(classroomId)
+getSchedulesByDay(day)
+addSchedule(schedule)
+updateSchedule(schedule)
+deleteSchedule(scheduleId)
manages